Customer Behaviour Analysis

Acomprehensive explorationofretaile-commerce data 4uncovering purchasepatterns, customer segments, discount effectiveness,and revenuedriversthroughPython EDA, SQLanalysis, andPowerBI visualization.

RETAIL / E-COMMERCE

STRUCTURED DATASET

CUSTOMER TRANSACTION LEVEL

About the Dataset

Dataset Overview

Type: Retail / E-commerce Dataset

Nature: Structured (Tabular)

Level: Customer Transaction Level

This dataset is used for **customer behavior analysis + business decision-making**

Data Mix

Numerical

Age,Purchase Amount, Previous Purchases

Categorical

Category,Payment Method, Season

Domain: E-Commerce Industry

- What customers buy
- How often they buy
- How much they spend
- Whether they respond to discounts

Business Problem & Key Questions

Howcan a retailcompanyunderstandcustomerpurchasingbehaviorto**increaserevenue,improve customer retention, and optimize marketingstrategies?**

Low Customer Retention

Who are frequent buyers vs. one-time buyers? Customers are not purchasing frequently enough.

Ineffective Marketing

Do discounts increase purchases? Which customers respond to promo codes?

Product Demand

Which category performs best? Which products generate the highest revenue?

Payment Behavior

Which payment methods are preferred? Are digital payments increasing?

Seasonal Demand

Does season impact sales? Which season drives maximum purchases?

Customer Segmentation

Identify high-value customers, frequent buyers, and discount-sensitive customers.

Data Dictionary

Eachrowinthedatasetrepresentsasinglecustomer transaction. Below is a complete reference of all columns used in the analysis.

Column Name	Description
Customer ID	Unique identifier for each customer
Age	Age of the customer
Gender	Gender of the customer (Male/Female/Other)
Item Purchased	Specific product purchased
Category	Product category (Clothing, Electronics, etc.)
Purchase Amount	Amount spent by the customer (USD)
Location	Customer's geographic location
Size	Product size (S, M, L, etc.)
Color	Product color
Season	Season of purchase (Winter, Summer, etc.)
Review Rating	Customer rating for the product
Subscription Status	Whether customer is subscribed (Yes/No)
Shipping Type	Type of shipping selected
Discount Applied	Whether discount was applied
Previous Purchases	Number of past purchases
Payment Method	Mode of payment
Frequency of Purchases	Purchase frequency (Weekly, Monthly, etc.)

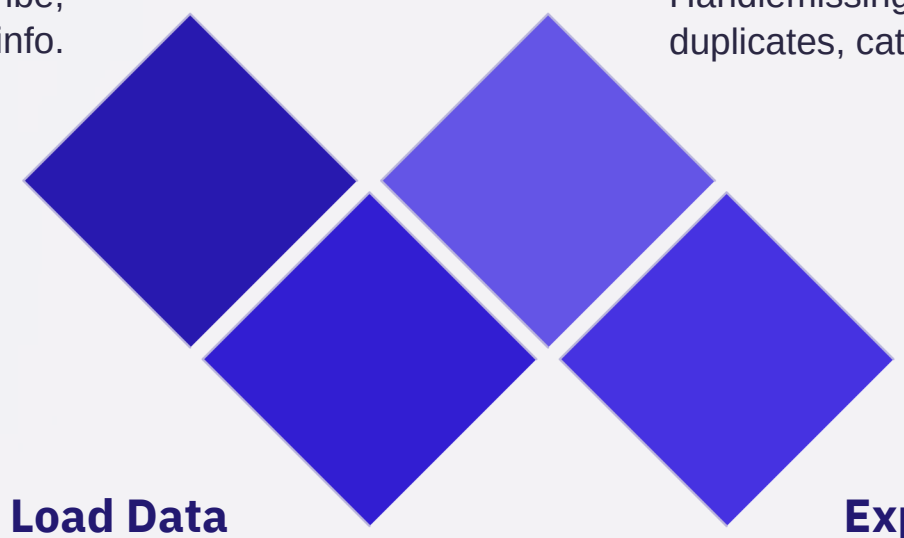
Exploratory Data Analysis (EDA) with Python

Explore Structure

Inspect with head, describe, and info.

Clean Data

Handlemissingvalues, duplicates, categories.



Load Data

Use pandas.read_csvto import CSV files.

Export to SQL

Writecleaned datavia sqlalchemy or pyodbc.

The EDA pipeline transforms raw CSV data into a clean, analysis-ready dataset exported to SQL Server for further querying and BI integration.

Data Loading

```
#importing the dataframe/dataset
df = pd.read_csv('customer_shopping_behavior.csv')
```

Brings raw data into a structured DataFrame, enabling further analysis and transformations.

Initial Exploration

- df.head() / df.tail() 4 view sample rows
- df.describe(include='all') 4 summary statistics
- df.info() 4 data types and structure
- df.isnull().sum() 4 detect missing values

Column Standardization

```
df.columns = df.columns.str.replace(" ", "_")
df.columns = df.columns.str.lower()
```

Makes column names clean, SQL-friendly, and easy to use in Power BI.

Export to SQL Server

```
pip install pyodbc sqlalchemy
```

Moves cleaned data into SQL for query-based insights and integration with BI tools.

Data Cleaning Strategy

Differentcolumnsrequireddifferentbusiness-logic-drivenimputation strategies 4 demonstrating real-world domain understanding.

Size Column 4 Category-Based Logic

- **Electronics / Accessories** ³ "Not Applicable" (no size concept)
- **Footwear** ³ "Free Size"
- **Clothing** ³ Filled with **mode** (most common size)

Review Rating 4 Product-Level Mean

Instead of the global average, the **product-level mean** was used. This keeps ratings realistic and meaningful 4 an advanced imputation technique.

Purchase Amount 4 Mean Replacement

Replaced with the mean value to maintain **product-level pricing consistency** and avoid distortion in revenue analysis.

Previous Purchases 4 Zero Fill

Missing values replaced with **0** 4 implying no previous purchase. Critical for customer segmentation and loyalty analysis.

Category Correction 4 Mapping Dictionary

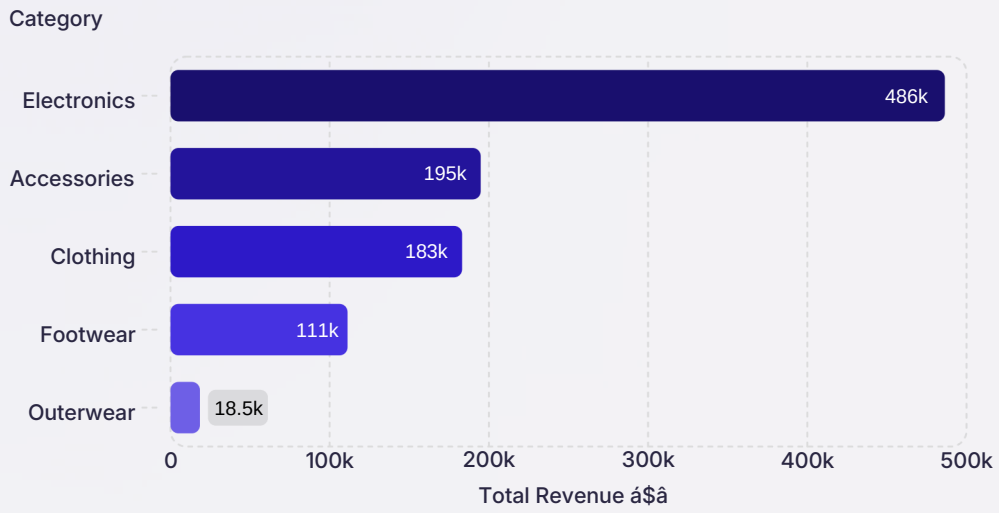
Found inconsistencies between Item Purchased and Category. Created a mapping dictionary to ensure **accurate grouping and correct business insights**.

Duplicate Handling 4 Customer ID

Identified and removed duplicates using Customer ID to ensure each customer appears only once. Critical for **accurate reporting**.

SQL Analysis 4 Revenue & Discount Insights

Revenue by Category

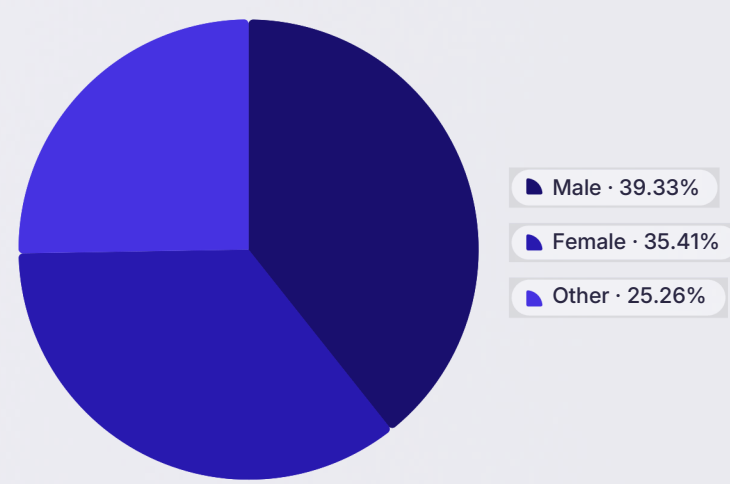


Electronics dominates revenue at $486K 4 nearly 2.5× the next category.

Discount Impact on Purchase Value

Customers who used discounts spent **$219.22 on average** vs. $182.24 without 4 a 20% uplift, confirming discounts drivehigher spend.

Revenue by Gender



SQL Analysis 4 Customers, Subscriptions & Segments

Subscriber vs. Non-Subscriber Spend

270.58

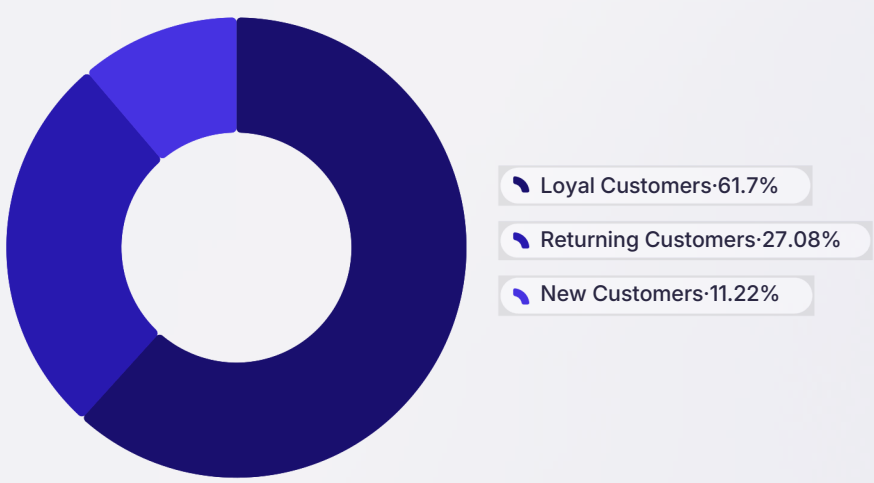
Average spend per customer (1,579 subscribers)

165.72

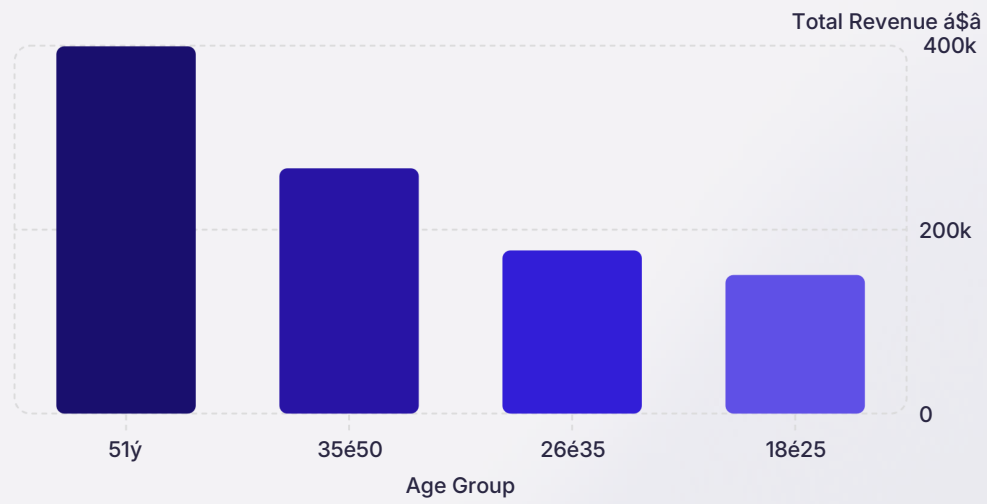
Average spendper customer (3,421 non-subscribers)

Subscribed customers spend **63% more** on average 4 a strongcase forgrowing thesubscription base.

Customer Segmentation



Revenue by Age Group



The **51+ age group** contributes 40.2% of total revenue 4 the most valuable demographicby far.

Repeat Buyers & Subscription Likelihood

Customer Type	Subscribed	Count
Repeat Buyers	Yes	1,190
Repeat Buyers	No	2,763
Normal Buyers	Yes	389
Normal Buyers	No	658

Power BI Dashboard

5000

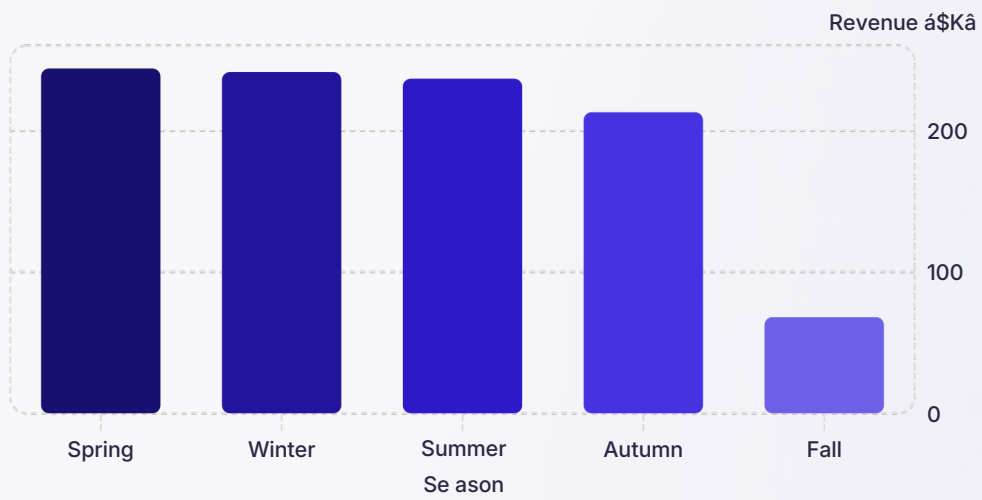
303.62 Unique Items Avg Rating

994K

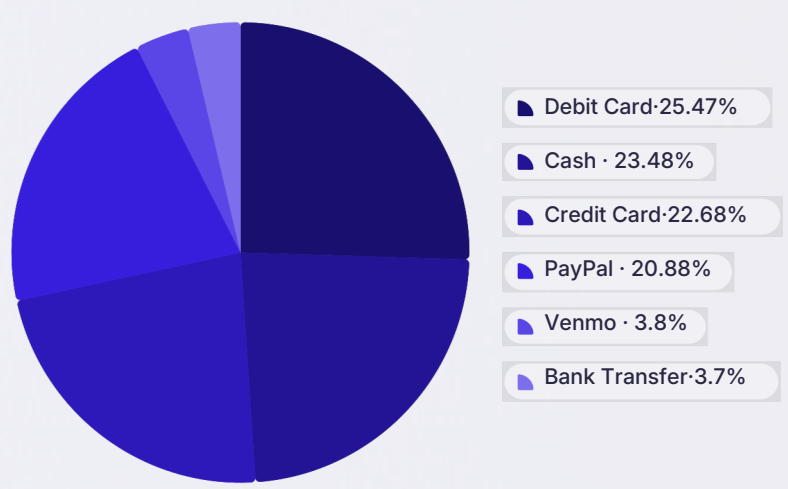
198.84

AvgSpend

Revenue by Season



Payment Method Distribution



Key Findings & Business Recommendations

Based on the fullanalysispipeline4fromPythonEDAthroughSQLqueriestoPowerBIdashboards4herearethe actionable insights for the business.

Electronics = #1 Revenue Driver

At $486K, Electronics generates nearly as much as all other categories combined. Prioritize inventory and marketing here.

Discounts Work 4 Strategically

Discounted customers spend 20% more on average. Focus promo codes on high-value segments to maximize ROI without eroding margins.

Subscribers Are High-Value

Subscribers spend $270 vs. $165 for non-subscribers. Growing the subscription base is a direct lever for revenue growth.

51+ Age Group Dominates

The 51+ cohort contributes 40.2% of total revenue. Tailor campaigns and product recommendations to this demographic.

Top Markets: NY, LA, Phoenix

New York ($143.9K), Los Angeles ($125.7K), and Phoenix ($119.1K) are the highest-revenue locations 4 ideal for geo-targeted campaigns.

Electronics Ratings Need Work

Phone (2.94), Headphones (3.08), and Watch (3.10) are the lowest-rated items despite being top revenue generators 4 a retention risk.

The full pipeline 4 Python EDA ³ SQL Analysis ³ Power BI Dashboard 4 delivers a 360° view of customer behavior, enabling data-driven decisions across marketing, product, and operations.